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	Title:		

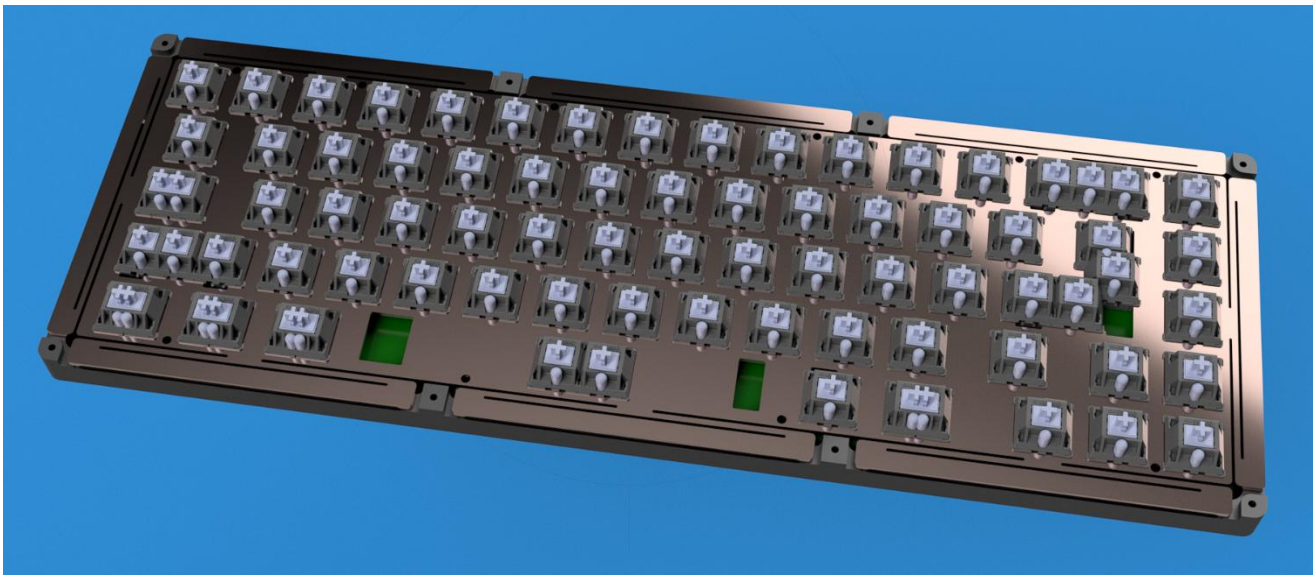
MONO65 Keyboard - electronics

Product Design Specification

Summary:

OPEN SOURCE

This document covers the design decisions behind the development phase of the project MONO65 Keyboard. Project consists of two boards:
- **MONO65** Keyboard (65% keyboard layout)
- **MONO65_DB** Daughter board (USB type-c to JST adapter board)



Keyboard render with MONO65_A1 PCB

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ABBREVIATIONS

Abbreviation	Description
PDS	Product Design Specification
MCU	Micro-controller Unit
PCB	Printed Circuit Board
QMK	Quantum Mechanical Keyboard
GND	Ground
I/O	Input and Output
ISP	In-System Programming
LED	Light Emitting Diode
MISO	Master In Slave Out
MOSI	Master Out Slave In
PCB	Printed Circuit Board
RST	Reset
SPI	Serial Peripheral/Programming Interface
XTAL	Crystal Oscillator
DFU	Device Firmware Upgrade
SCK	System Clock

USEFUL LINKS

Link	Description
https://salvun.com/collections/store/products/wilba-graphite-pcb	Salvun PCB 65% as an inspiration project
https://github.com/ai03-2725/MX_Alps_Hybrid	MX Cherry GitHub library
https://docs.qmk.fm/#/	QMK Firmware
https://github.com/qmk/qmk_firmware	QMK Firmware GitHub repo
https://wiki.ai03.com/books/pcb-design/chapter/pcb-designer-guide	PCB Design guide
https://github.com/help-14/mechanical-keyboard	Mech. Keyboards GitHub repo overview
https://cdn.sparkfun.com/datasheets/Components/Switches/MX%20Series.pdf	MX Cherry Switch Datasheet
www.youtube.com/watch?v=vLGklanzQIc	The Switch Matrix - PCB Design for Mechanical Keyboards Pt. 3
https://github.com/qmk/qmk_firmware/blob/master/docs/feature_backlight.md	QMK Backlight tutorial
www.microchip.com/en-us/product/AT90USB1286	AT90USB1286-MU – selected 8-bit AVR micro-controller from Microchip (Atmel) 64-pin QFN
https://github.com/ai03-2725/Unified-Daughterboard	Unified Daughter board Project – GitHub repo
https://www.microchip.com/en-us/development-tool/ATATMEL-ICE	ATMEL ICE programmer for AVR
https://www.tag-connect.com/debugger-cable-selection-installation-instructions/atmel-ice#142_122_148	Tag connect cable - ISP programming for AVR
https://www.mouser.com/datasheet/2/268/doc7627-1180783.pdf	AT90USBKey Hardware User Guide
https://ww1.microchip.com/downloads/en/DeviceDoc/doc7618.pdf	USB DFU Bootloader Datasheet
https://www.microchip.com/en-us/development-tool/flip	Microchip Flip tool
https://led-ld.nichia.co.jp/api/data/spec/led/NSPW310DS-E(1461).pdf	3mm White LED Nichia SPW310DS Datasheet
https://github.com/ai03-2725/Unified-Daughterboard	Unified Daughterboard Project Github repo

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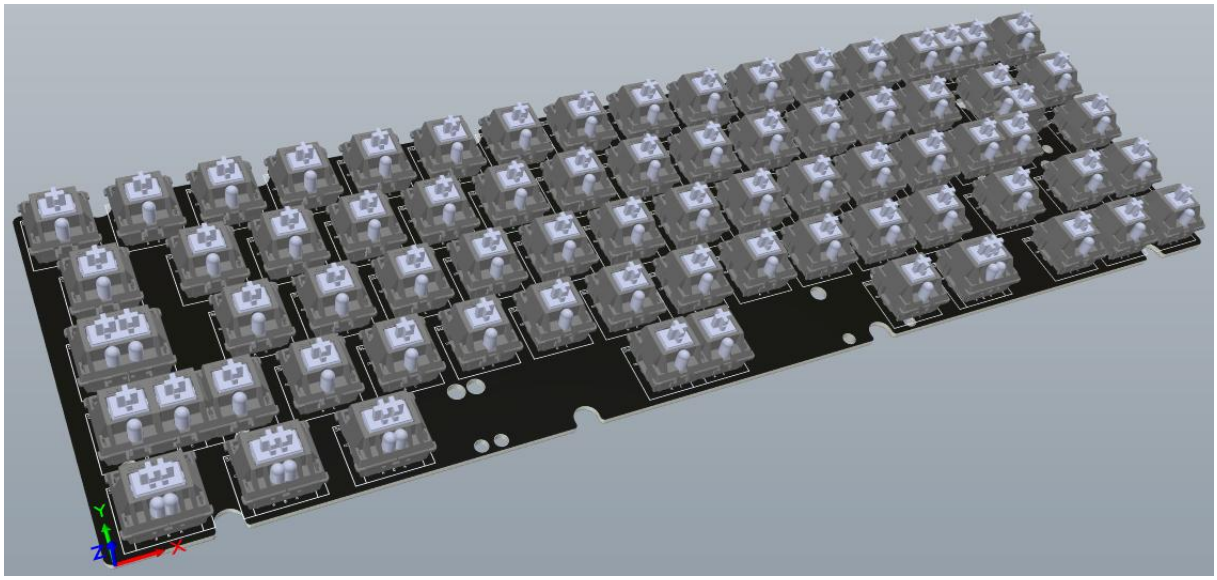
1. INTRODUCTION

This document covers the design decisions behind the development phase of the project MONO65 Keyboard. Project consists of two boards:

- MONO65 Keyboard (65% keyboard layout)
- Daughter board (USB type-c to JST adapter board)

1.1. Project overview

MONO65 is an unofficial name for the keyboard in 65% format designed with Tomaz Zaman. The keyboards' electronics is placed in the custom milled aluminum enclosure. The keyboard is supported by all major Operating Systems; Windows, Linux and Apple OS. Its core is Atmel MCU AT90USB1286-MU with QMK firmware. On the computer side it is mandatory to use Via Configuration software to properly program (map) all the buttons.

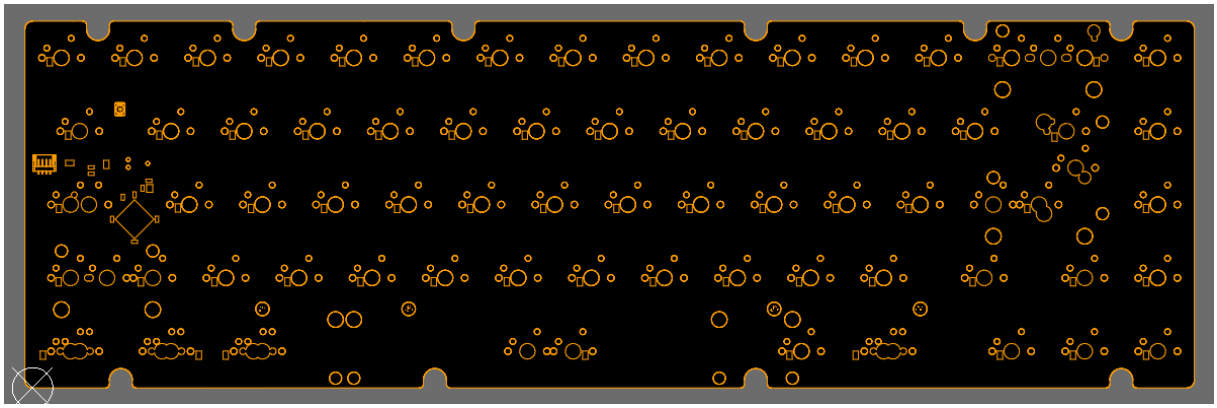


MONO65_A1 Keyboard PCB mechanical concept

2. MONO65_A1 Keyboard

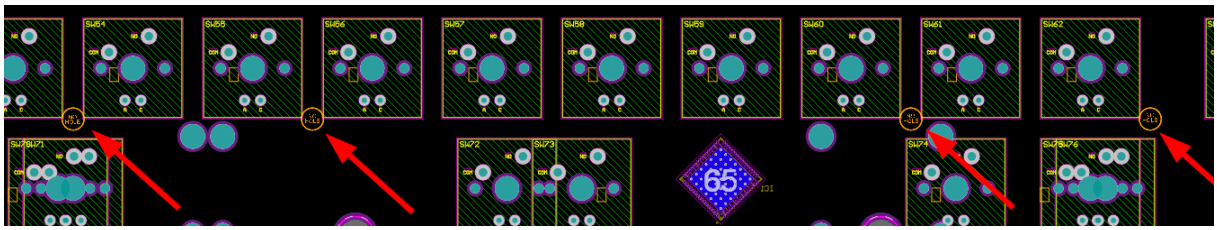
2.1. Mechanics and dimensions

The board's shape was inspired by the Wilba.tech's keyboard.



MONO65_A1 Keyboard – Altium Designer Mechanical Layer 100

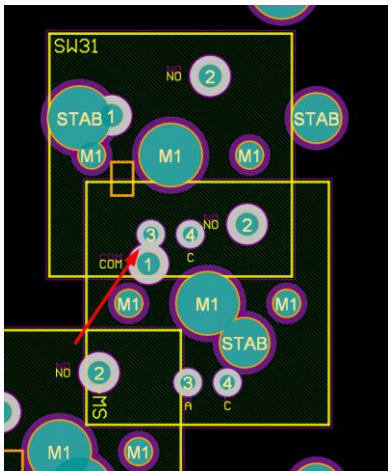
Additionally, we decided to skip 4 holes from the reference mechanical drawing. These holes are supposed to be used by another configuration, which is not supported by the MONO65 keyboard.



MONO65_A1 Keyboard – Skipped mechanical holes

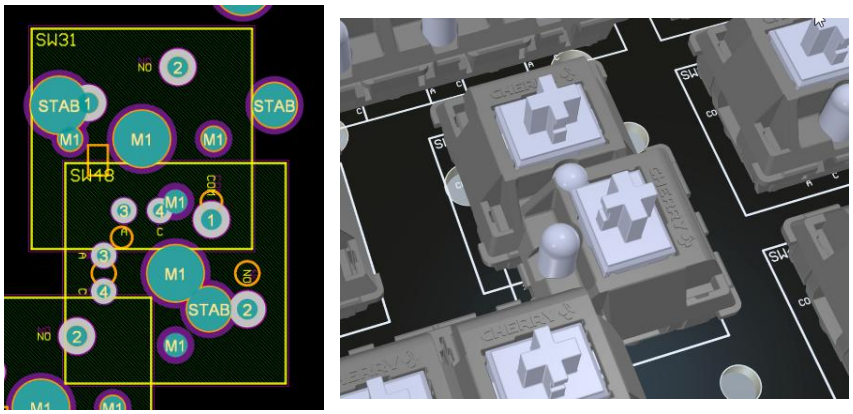
2.2. Switches placement

Switches were placed according to chapter 2.1., and their position was verified by Tomaz. The difference between our design and Wilba.tech's is in the switches type. Wilba.tech's board doesn't support LEDs and MONO65 Keyboard does support single color THT LEDs. The difference is causing the problem with overlapping some pins which must not be electrically connected to each other – drawings below.



ONO65_A1 Keyboard – Switches SW31 & SW48 pins 1 and 3 are shorted

This issue was bypassed by rotating the switch SW48 clockwise for 90 degrees – photo below. The orange layer (Mechanical layer 100) shows the previous (Wilba.tech's) orientation. With this orientation there are no shorts between electrical pins. Mechanical layer 100 still consists of Wilba.tech's drawings.

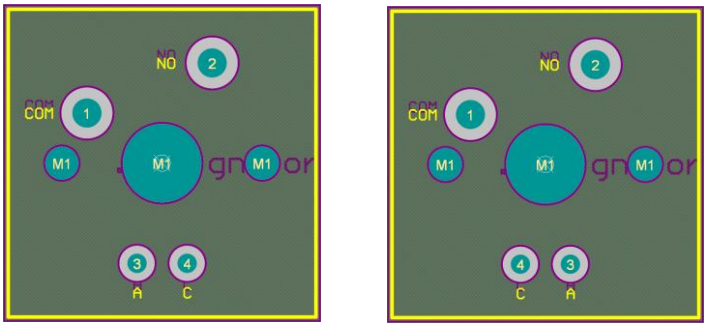


MONO65_A1 Keyboard – Switch SW48 rotated for 90 degrees clockwise

A similar issue occurred for all switches which are partially overlapping with the center-center distance of 2.381mm where pins for LED diodes are shorted. Switches with ref designators SW66, SW67, SW68, SW69, SW70, SW71, SW75 and SW76 are affected. This issue was fixed by introducing another MX cherry footprint variant where pins for LED were inverted.

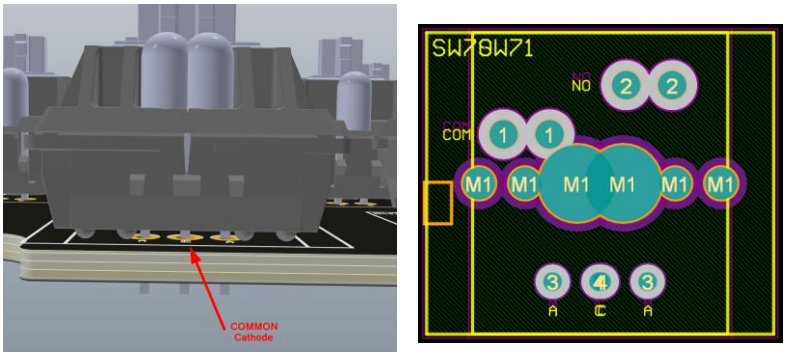
Switches with the footprint "MX1A-1-reversed LED-Pins": SW67, SW69, SW71 and SW76.

Switches with the footprint "MX1A-1-LED-Pins": all others.



MONO65_A1 Keyboard – MX Cherry Switch library comparison. Left default lib - "MX1A-1-LED-Pins". Right new lib variant with the inverted LED - "MX1A-1-reversed LED-Pins"

This solution contributed to the intentional reuse of cathode pins. The photo below is showing the overlapping pins in 2D and 3D views. Additional instructions are needed on a silkscreen layer to prevent mistakes during the



assembly.

MONO65_A1 Keyboard – MX Cherry Switch library comparison. Left LED inverted. Right default lib

2.3. Board Stackup and technology

The board was designed with production costs in mind and was optimized for 2 layer stackup, total thickness 1.6mm. The priority was put on the surface's finish instead, with the goal to achieve a unique end-product look.

Technology specifications:

- Material: Tg140 FR-4 BLACK

- Thickness: 1.6mm

- Min Track/Spacing: 8/8mils or ~0.2/0.2mm

- Min Hole Size: 0.2mm

- Solder mask: Transparent

- Silkscreen: Grey

- Surface Finish: ENIG

- Finished Copper: 1 oz Cu

- No UL marking

2.4. Atmel MCU

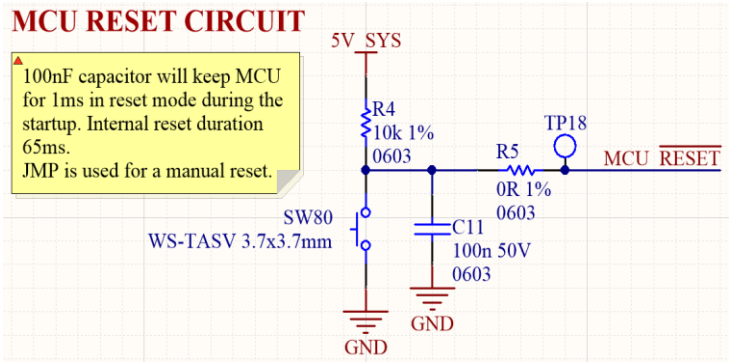
For the MONO65 keyboard project the AT90USB1286-MU, 8-bit AVR MCU from Microchip (Atmel) in 64-pin QFN package was selected. This MCU is well supported by the VIA application and offers enough pins to support higher number of switches – full size keyboard.

The 64-pin QFN package is in size of 9.00mm x 9.00mm with 0.5mm pitch size.

2.4.1. Reset

Reset input pin: A low level on this pin for longer than the minimum pulse length will generate a reset, even if the clock is not running.

MCU reset pin is pulled up with 10k resistor, which holds MCU out of the reset mode during the normal operation. The button switch SW80 is used to put the processor in the reset mode while pressing by a finger. During the startup, the MCU is kept in the reset mode for 65ms due to the internal reset signal, after that if the external reset signal is high, the MCU will jump into the normal working mode.



MONO65_A1 Keyboard – MCU reset line

2.4.2. MCU Clock

2.4.2.1. Clock sources

The device has the following clock source options, selectable by Flash Fuse bits as shown below. The clock from the selected source is input to the AVR clock generator and routed to the appropriate modules. MONO65 keyboard was designed to use an external Low power crystal oscillator which must be enabled by Flash Fuse bits.

Table 7-1. Device clocking options select (1).

Device clocking option	CKSEL3..0
Low power crystal oscillator	1111 - 1000
Reserved	0111 - 0110
Low frequency crystal oscillator	0101 - 0100
Reserved	0011
Calibrated internal RC oscillator	0010
External clock	0000
Reserved	0001

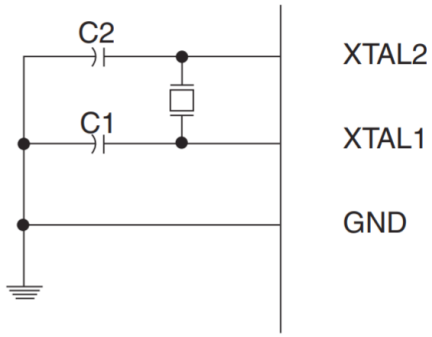
Note: 1. For all fuses "1" means unprogrammed while "0" means programmed.

MONO65_A1 Keyboard – MCU clock sources datasheet guideline – page 41

2.4.2.2. Low power crystal oscillator

Pins XTAL1 and XTAL2 are input and output, respectively, of an inverting amplifier which can be configured for use as an On-chip Oscillator, as shown in Figure below. Either a quartz crystal or a ceramic resonator may be used. This Crystal Oscillator is a low power oscillator, with reduced voltage swing on the XTAL2 output. It gives the lowest power consumption, but is not capable of driving other clock inputs, and may be more susceptible to noise in noisy environments.

Figure 7-2. Crystal oscillator connections.



MONO65_A1 Keyboard – MCU crystal oscillator connection datasheet guideline – page 43

The low power oscillator can operate in three different modes, each optimized for a specific frequency range. The operating mode is selected by the fuses CKSEL3..1 as shown below. The CKSEL3..1 "111" shall be used for this application.

Table 7-3. Low power crystal oscillator operating modes [3].

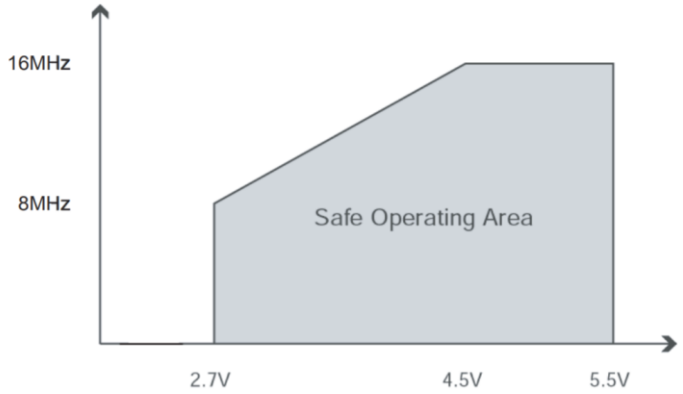
Frequency range [1] [MHz]	CKSEL3..1	Recommended range for capacitors C1 and C2 [pF]
0.4 - 0.9	100 [2]	–
0.9 - 3.0	101	12 - 22
3.0 - 8.0	110	12 - 22
8.0 - 16.0	111	12 - 22

Notes:
1. The frequency ranges are preliminary values. Actual values are TBD.
2. This option should not be used with crystals, only with ceramic resonators.
3. If 8MHz frequency exceeds the specification of the device (depends on V_{CC}), the CKDIV8 Fuse can be programmed in order to divide the internal frequency by 8. It must be ensured that the resulting divided clock meets the frequency specification of the device.

MONO65_A1 Keyboard – Low power crystal oscillator operating modes datasheet guideline – page 43

Maximum frequency depends on VCC. As shown in Figure bellow (datasheet 31-2 on page 393), the maximum frequency vs. VCC curve is linear between 2.7V < VCC < 5.5V. The keyboard is powered via an USB "5V_SYS" which allows the frequency to be high as 16MHz.

Figure 31-2. Maximum frequency vs. V_{CC}, Atmel AT90USB64/128.

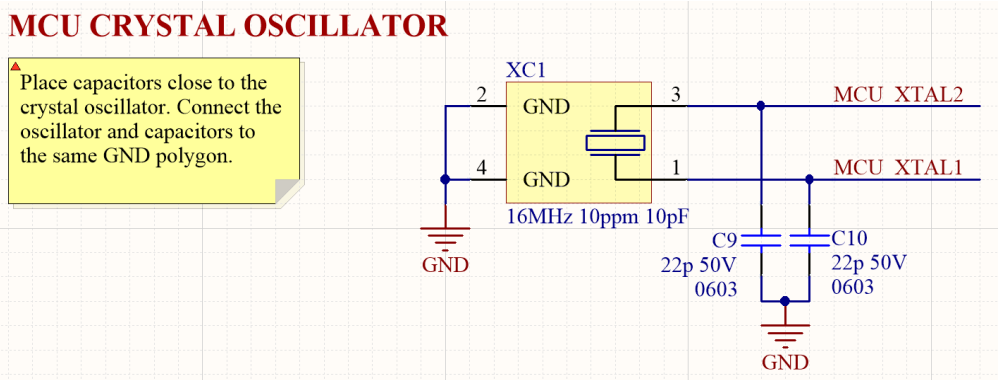


MONO65_A1 Keyboard – Maximum frequency vs. VCC supply datasheet guideline – page 393

The figure below shows the connection between the crystal oscillator and MCU.

Selected 16MHz 10ppm 10pF crystal oscillator (XC1) is MCSJK-7U-16.00-10-10-80-B-10 from Multicomp in SMD 3225 package.

Capacitors C9 and C10 are 22pF 50V C0G/NP0 in 0603 package size and are placed close to the crystal itself.



MONO65_A1 Keyboard – 16MHz crystal oscillator

2.4.2.3. USB oscillator 48MHz

USB 48MHz clock is internally generated via a PLL (2MHz → 48MHz). No need for an external clock.

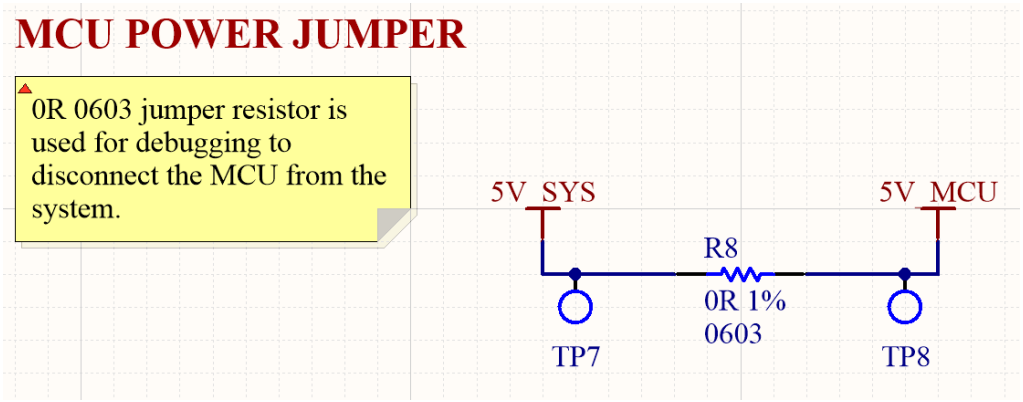
7.1.6 USB Clock – clk_{USB}

The USB is provided with a dedicated clock domain. This clock is generated with an on-chip PLL running at 48MHz. The PLL always multiply its input frequency by 24. Thus the PLL clock register should be programmed by software to generate a 2MHz clock on the PLL input.

MONO65_A1 Keyboard – 48MHz USB clock datasheet guideline

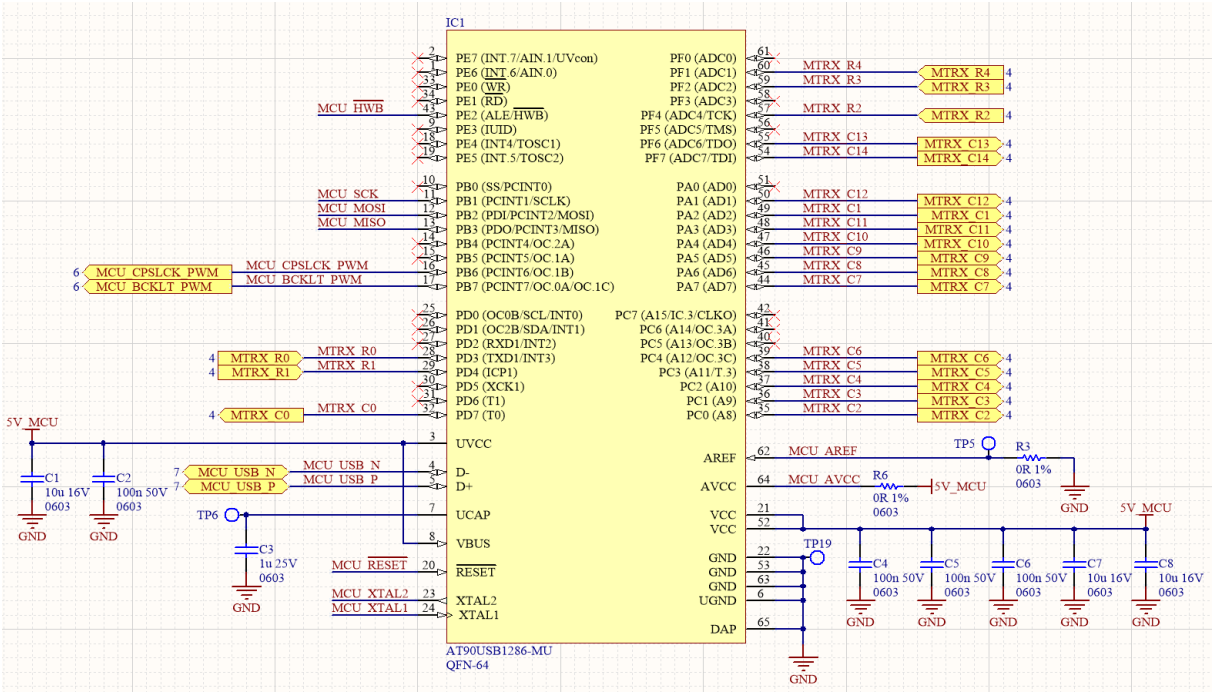
2.4.3. Powering the MCU

The MCU is powered from the USB power line – 5V and its power line can be interrupted by removing the jumper 0R resistor. The rationale behind using the jumper is to have a possibility of disconnecting the MCU from the system during the troubleshooting or to insert an Ampere-meter in series between the load and its source.



MONO65_A1 Keyboard – R8 0R Jumper resistor on the MCU power line 5V

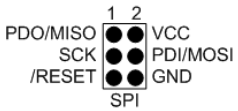
According to the datasheet the bypass capacitors are used for each power pin on the micro-controller. The schematic below shows the



2.4.4. Programming the MCU

The MCU can be programmed via an ISP (In-System Programming) port which utilizes the SPI bus.

In-System Programming uses the target Atmel AVR's internal SPI (Serial Peripheral Interface) to download code into the flash and EEPROM memories. **It is not a debugging interface.** When designing an application PCB, which includes an AVR with the SPI interface, the pinout as shown in the figure below should be used.



MONO65_A1 Keyboard – SPI interface

The programmer unit, suitable for programming the MCU is ATMEL-ICE which supports all AVR microcontrollers. Such programmer offers a 10-pin connection interface where a flat cable can be plugged. The Tag Connect TC2030-ICESPI-NL No Leg Cable for use with Atmel-ICE is visible in the photo below.

Note: The 10-pin male connectors on the Atmel-ICE have reversed (non-standard) pin numbering. For Atmel-ICE you should purchase the LEMTA version of our cables which have the 10-pin ribbon connector fitted backwards which corrects for the Atmel-ICE pin numbering reversal.



MONO65_A1 Keyboard – SPI interface

Below tables show the pinning configuration for the 6-pin Tag Connect and 10-pin pinheader connectors.

TC2030-ICESPI-NL Connections

TC2030	SPI	PDI	aWire	10-pin IDC
1	MISO	PDI_DATA	DATA	3 TDO
2	VTGS	VTref	VTG	4 VTG
3	SCK			1 TCK
4	MOSI			9 TDI
5	/RESET	PDI_CLK		6 nSRST
6	GND	GND	GND	2 GND

MONO65_A1 Keyboard – TC2030 SPI pinning (Target PCB side)

10-Pin IDC	LEMTA IDC	SPI	TPI	PDI	aWire	debugWIRE	6-pin TC2030
1 TCK	10	SCK	SCK				3
2 GND	9	GND	GND	GND	GND	GND	6
3 TDO	8	MISO	DATA	PDI_DATA	DATA		1
4 VTG	7	VTG	VTG	VTG	VTG	VTG	2
5 TMS	6						
6 nSRST	5	/RESET	/RESET	PDI_CLK		RESET	5
7 n/c	4						
8 nTRST	3						
9 TDI	2	MOSI					4
10 GND	1						

MONO65_A1 Keyboard – TC2030 SPI (IDC) pinning (Atmel-ICE side)

Programming with USB bootloader: DFU (Device Firmware Upgrade)

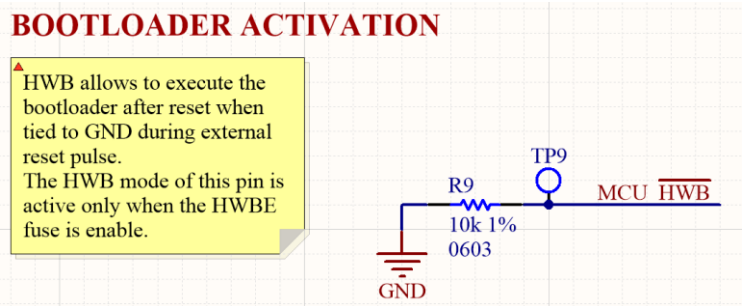
AT90USB device comes with a default factory pre-programmed USB bootloader located in the on-chip boot section of the AT90USB. This is the easiest and fastest way to reprogram the device directly over the USB interface. The “Flip” PC side application, is available from the Atmel website, offers a flexible and user friendly interface to reprogram the application over the USB bus. The HWB pin of the AT90USB allows to force the bootloader section execution after reset. (Refer to AT90USB datasheet section “boot loader support”).

To force bootloader execution, operate as follows:

1. Press “RESET SW80” (HWB line is already pulled low by default)
2. Release the “RESET SW80” push button
3. HWB mode activated

For more information about the USB bootloader and “Flip” application, please refer to the “USB bootloader datasheet document” and “Flip’s user manual”. Note: The HWB pin is active only if the HWBE fuse is set (default factory configuration).

During the system startup, the internal reset is going to hold the MCU in a reset mode for 65ms after that if the external reset line is pulled high, the MCU will jump into a normal working mode. After that, if the Reset button is pressed the MCU will jump into Bootloader mode, due to the HWB pin being pulled down with an external resistor.



MONO65_A1 Keyboard – Hardware Bootloader pulled down by default

- **ALE/HWB – Port E, bit 2**

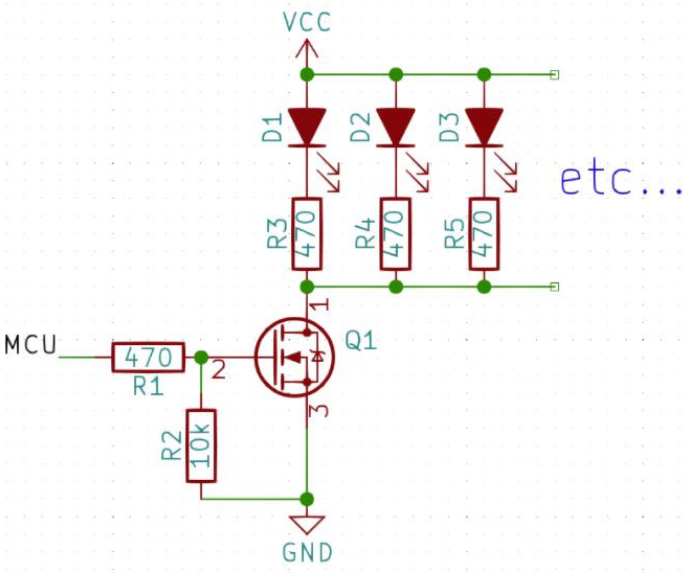
ALE is the external data memory Address latch enable.

HWB allows to execute the boot loader section after reset when tied to ground during external reset pulse. The HWB mode of this pin is active only when the HWBE fuse is enable.

MONO65_A1 Keyboard – Hardware Bootloader fuse HWBE

2.5. LED backlight

The MONO65 keyboard offers backlighting for each button. The latter is controlled via a PWM signal which comes from the MCU port PB7. Additionally, the CapsLock switch has its own PWM controlled LED which is connected to the PB6 of the MCU. The LEDs are wired according to the QMK tutorial recommendations – in parallel, with a current limiting resistor in series to each LED. The schematic below shows the basic concept of wiring the LEDs.



MONO65_A1 Keyboard – QMK tutorial of wiring the LED backlight to a single MCU's pin

LED backlight resistor was calculated for Nichia NSPW310DS LEDs.

https://led-ld.nichia.co.jp/api/data/spec/led/NSPW310DS-E(1461).pdf

SPECIFICATIONS

(1) Absolute Maximum Ratings

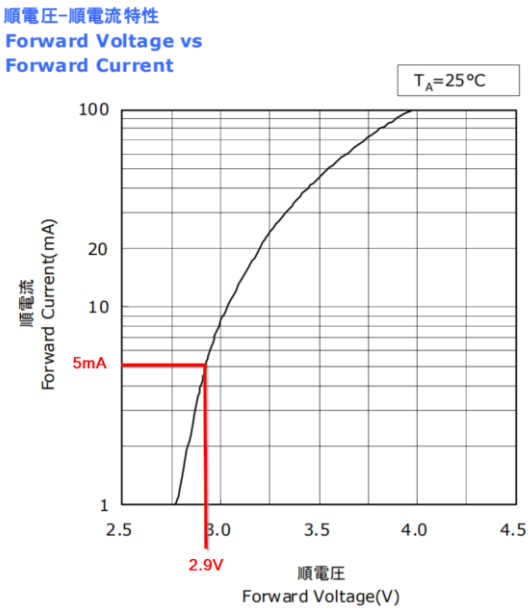
Item	Symbol	Absolute Maximum Rating	Unit
Forward Current	I_F	30	mA
Pulse Forward Current	I_{FP}	100	mA
Reverse Voltage	V_R	5	V
Power Dissipation	P_D	105	mW
Operating Temperature	T_{opr}	-30~85	°C
Storage Temperature	T_{stg}	-40~100	°C
Junction Temperature	T_J	100	°C

* Absolute Maximum Ratings at T_A=25°C.
* I_{FP} conditions with pulse width ≤10ms and duty cycle ≤10%.

(2) Initial Electrical/Optical Characteristics

Item		Symbol	Condition	Typ	Unit
Forward Voltage		V_F	I_F=20mA	3.2	V
Luminous Intensity		I_v	I_F=20mA	6.2	cd
Chromaticity Coordinate	x	-	I_F=20mA	0.31	-
	y	-	I_F=20mA	0.32	-

MONO65_A1 Keyboard – Nichia NSPW310DS LED specifications



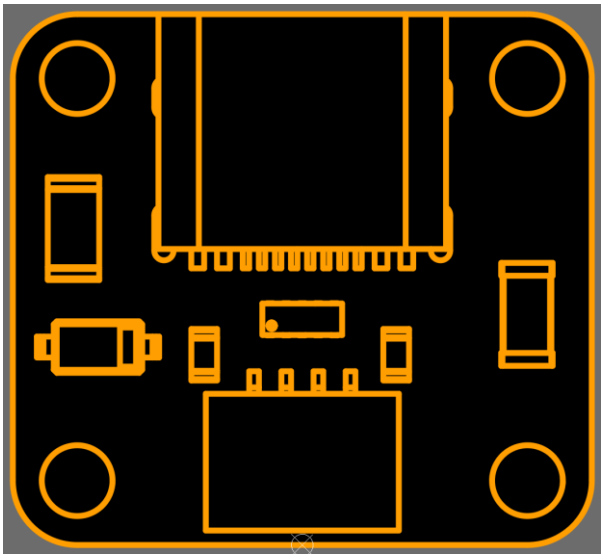
MONO65_A1 Keyboard – Nichia NSPW310DS LED Voltage drop over the LED @5mA

2.6. Protections

3. MONO65_DB_A1 Daughter board

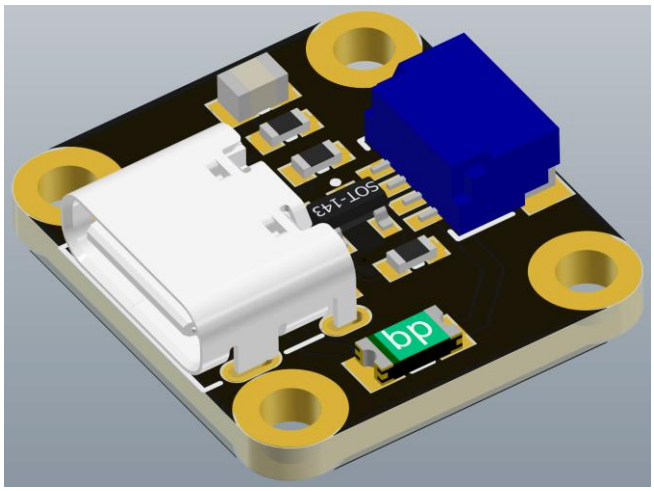
3.1. Mechanics and dimensions

The shape of the daughter board was defined with help of an open source UNIFIED C3 board. It's step model was converted with SolidWorks into the DXF and imported into Altium Designer PCB file – Mechanical layer 100. On that layer the board's shape and connectors placement are visible.



MONO65_DB_A1 Daughter board – Altium Designer Mechanical Layer 100

Additionally, we decided to electrically connect all 4 mounting holes with the enclosure. The MNT1, MNT2, MNT3 and MNT4 are M2 plated holes with an inner diameter of 2.2mm. The enclosure should provide good electrical contact with at least one mounting hole. To achieve sufficient conductivity of the enclosure, the anodized contacting surfaces shall be CNC milled away.



MONO65_DB_A1 Daughter board – Altium Designer 3D view – top side

3.2. Protections

3.3. Board Stackup and technology

The board was designed with production costs in mind and was optimized for 2 layer stackup, total thickness 1.6mm. The priority was put on the surface'a finish instead, with the goal to achieve a unique end-product look.

Technology specifications:

- Material: Tg140 FR-4 BLACK

- Thickness: 1.6mm

- Min Track/Spacing: 5/5mils or ~0.125/0.125mm

- Min Hole Size: 0.25mm

- Solder mask: Transparent

- Silkscreen: Grey

- Surface Finish: ENIG

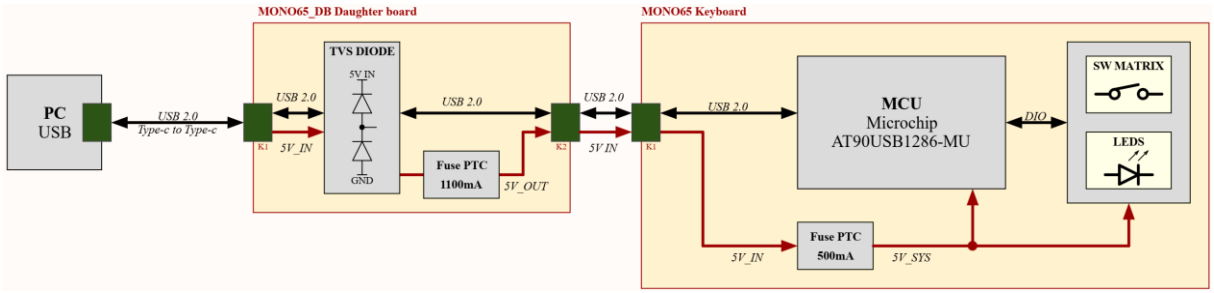
- Finished Copper: 1 oz Cu

- No UL marking

4. B2B INTERCONNECTION

4.1. Overview

This chapter covers an overview of the board to board connectors and cabling used for connecting the keyboard and the daughter board together.



MONO65_A1 Connectivity – Altium Designer Mechanical Layer 100

JST SH connector series 1mm pitch

Current rating 1.0A @ 28AWG AC/DC

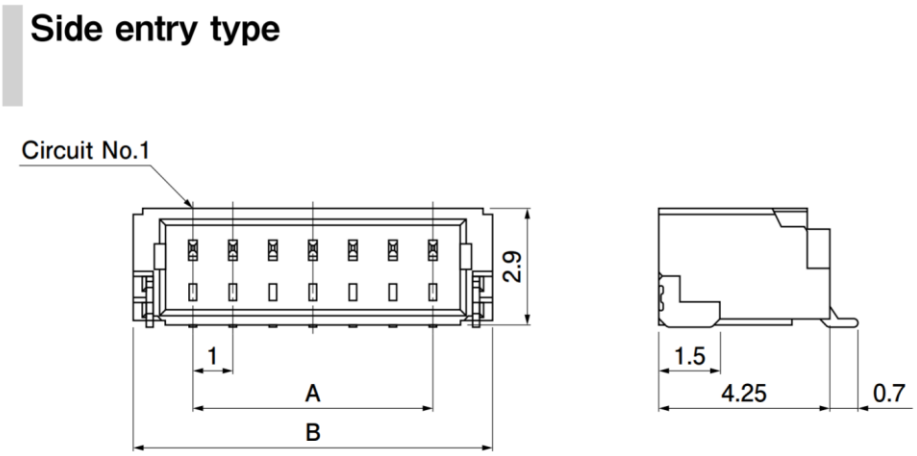
Voltage rating 50V AC/DC

Applicable wire: Conductor size 28-32AWG

4.2. Connectors

4.2.1. JST SM04B-SRSS-TB (PCB)

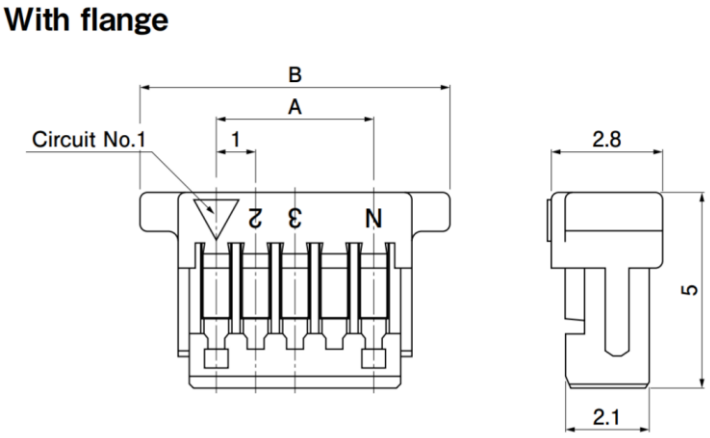
PCB side: Size entry SM04B-SRSS-TB



MONO65_A1 Connectivity – JST SSH-003T-P0.2-H crimp contact

4.2.2. JST SHR-04V-S-B (Plug)

Wire side: SHR-04V-S-B

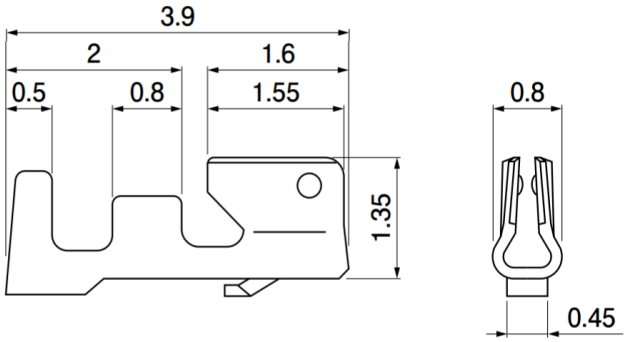


MONO65_A1 Connectivity – JST SSH-003T-P0.2-H housing w/ flange

4.2.3. JST SSH-003T-P0.2-H (Crimp)

Crimp contact: SSH-003T-P0.2-H

Applicable wire: 0.032-0.08 mm², 28-32AWG, insulation outer diameter 0.4-0.8mm



MONO65_A1 Connectivity – JST SSH-003T-P0.2-H crimp contact

4.2.4. MOLEX USB TYPE-C REC (PCB)

4.3. JST-JST CABLE